RGUKT-R K Valley Campus

AY2025-26: Sem-I	Engg-IV	MID-1	R20 B.Tech – CSE
Subject Code: 20CS4101	Subject Name: Machine Learning		
Time: 60 Min	Max.Marks:20	Date of Exam: 7[redacted]-Aug-2025	

Part –A (Answer all questions, each question carries 1 Mark) | 8 x 1 M = 8 Marks

1.	Which of the following best defines Machine Learning?			
	A)	A technique to design algorithms manually	B)	A method to store and retrieve large datasets
	C)	A system that improves its performance with experience	D)	A set of programming rules for hardware interaction
2.	The ability of a model to perform well on unseen data is called:			
	A)	Underfitting	B)	Generalization
	C)	Memorization	D)	Classification
3.	Version Space is defined as:			
	A)	All possible hypotheses	B)	Set of hypotheses consistent with all training examples
	C)	Union of all attribute values	D)	Space of all testing errors
4.	A drawback of decision trees is:			
	A)	They handle missing data easily	B)	They are resistant to overfitting
	C)	They require normalization	D)	They are prone to overfitting in noisy data
5.	In KNN, the value of K affects:			
	A)	Bias-variance tradeoff	B)	Feature scaling
	C)	Model complexity only	D)	Class labels
6.	The bandwidth parameter τ in LWR controls:			
	A)	The shape of the hypothesis	B)	The speed of convergence
	C)	The weight decay with distance	D)	The number of neighbors
7.	The Gaussian RBF function uses which component to control spread?			
	A)	Learning rate	B)	Activation threshold
	C)	Bandwidth (σ or τ)	D)	Epoch count
8.	Instance-based learning stores:			
	A)	Only weights of the model	B)	Only the training data
	C)	A function approximation	D)	Predefined class labels

Part –B (Answer all questions, each question carries 6 Mark) **2 x 6 M = 12 Marks**

Scientific Calculator may be permitted.

1. **A.** Explain the types of Machine Learning with neat example. **[4M]**

   B. Find the most Specific and General boundaries of below data. **[2M]**

Sky	Temperature	Humidity	Wind	Water	Forecast	Play Cricket
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cold	Change	Yes

(OR)

2. Explain the Decision Tree Learning. And Construct the Decision Tree for below dataset. [6M]

ID	Credit Score	Income Level	Collateral	Loan Approved
1	Good	High	Yes	Yes
2	Good	High	No	Yes
3	Good	Low	Yes	No
4	Bad	High	Yes	Yes
5	Bad	Medium	No	No
6	Bad	Low	Yes	No
7	Good	Medium	Yes	Yes
8	Good	Medium	No	No
9	Bad	Medium	Yes	Yes
10	Bad	High	No	No

3. Explain the K-Nearest Neighbours algorithm. And classify the colour if Brightness = 75 and Saturation = 60 for the below data. [6M]

ID	Brightness (0–100)	Saturation (0–100)	Class
1	85	75	Red
2	30	90	Blue
3	70	40	Green
4	90	80	Red
5	25	85	Blue
6	60	35	Green
7	80	70	Red
8	28	95	Blue
9	65	30	Green

(OR)

4. **A.** Explain the Radial basis function. and predict the price for size = 1100sq. for below data.[4M]

Size (x)	Price (y)
800	150
1000	200
1200	240
1500	300

B. Explain the Locally Weighted Regression briefly. [2M]

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AY2025-26: S-I Engg-IV MID-2 R20 B.Tech – CSE

20CS4101 Machine Learning

Time: 60 Min Max.Marks:20 Date: 27-Sept-2025

Part A (Answer all questions, each question carries 1 M) 8 x 1 M = 8 M

1. A perceptron is _____
   - A) a single layer feed-forward neural network with pre-processing.
   - B) an auto-associative neural network
   - C) a double layer auto-associative neural network
   - D) a neural network that contains feedback
2. In a multilayer perceptron, the backpropagation algorithm is primarily used for
   - A) Feature selection
   - B) Weight adjustment
   - C) Data normalization
   - D) Dimensionality reduction
3. Genetic algorithms are inspired by:
   - A) Human learning theories
   - B) Natural evolution and genetics
   - C) Quantum computing
   - D) Statistical sampling
4. Which activation function is most commonly used in backpropagation networks?
   - A) Step function
   - B) Sigmoid function
   - C) Linear function
   - D) None of these
5. The main goal of Support Vector Machines (SVM) is to:
   - A) Minimize classification accuracy
   - B) Find the hyperplane with the maximum margin
   - C) Reduce dimensionality
   - D) Perform clustering
6. Which kernel is commonly used in SVM to handle non-linear classification problems?
   - A) Linear kernel
   - B) Radial Basis Function (RBF) kernel
   - C) Polynomial kernel
   - D) Both B and C
7. In Reinforcement Learning, the term "policy" refers to:
   - A) The agent's strategy of selecting actions
   - B) A set of training data
   - C) A performance metric
   - D) The reward function
8. Q-learning is a type of:
   - A) Supervised learning
   - B) Unsupervised learning
   - C) Model-free reinforcement learning
   - D) Genetic learning

Part –B (Answer all questions, each question carries 6 M) 2 x 6 M = 12 M

Scientific Calculator may be permitted.

1. Explain the architecture of a multilayer neural network. How does backpropagation update the weights?

(OR)

2. Describe the genetic algorithm process with an example. How do crossover and mutation help in optimization?

3. A. Describe the working behavior of support vector machine with neat diagrams.

   B. Explain the types of Support Vector Machine in Machine Learning

(OR)

4. Explain the working of Q-learning with an example. How does it differ from model-based approaches?

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AY 2025-26: Sem-I Engg-IV MID-3 R20 B.Tech – CSE

Subject Code: 20CS4101 Subject Name: Machine Learning

Time: 60 Min Max.Marks:20 Date of Exam: 7-Nov-2025

Part –A (Answer all questions, each question carries 1 M) 8 x 1 M = 8 Marks

1. In Bayesian learning, the hypothesis with the maximum posterior probability is called:
   - A) Bayes Optimal Hypothesis
   - B) Maximum Likelihood Hypothesis
   - C) Minimum Description Length Hypothesis
   - D) Naïve Hypothesis
2. Maximum Likelihood Estimation assumes:
   - A) Uniform prior distribution
   - B) Data is independent of model
   - C) Posterior equals likelihood
   - D) The hypothesis that makes observed data most probable
3. In Naïve Bayes Classifier, the "naïve" assumption refers to:
   - A) Prior probabilities are zero
   - B) Data contains no noise
   - C) Only one attribute affects the output
   - D) All attributes are independent given the class
4. The Minimum Description Length (MDL) principle is based on:
   - A) Shortest encoding of hypothesis and data
   - B) Maximum entropy
   - C) Least squares error
   - D) Conditional probability only
5. Sample complexity refers to:
   - A) The number of hidden layers
   - B) The complexity of computation
   - C) The number of samples needed to learn a hypothesis
   - D) The hypothesis length
6. In mistake bound model, the bound refers to:
   - A) Number of samples used
   - B) Maximum training time
   - C) Number of errors before convergence
   - D) Margin width
7. Sequential covering algorithm is used in:
   - A) Rule-based learning
   - B) Neural networks
   - C) SVM optimization
   - D) Reinforcement learning
8. First-order rules differ from propositional rules because they:
   - A) Use variables and quantifiers
   - B) Use only constants
   - C) Are based on probability distributions
   - D) Exclude relational data

Part –B (Answer all questions, each question carries 6 M) 2 x 6 M = 12 Marks

Scientific Calculator may be permitted.

1. Given two hypotheses H_1 and H_2 with likelihoods 0.6 and 0.8, and priors $P(H_1) = 0.7$, $P(H_2) = 0.3$, compute normalized posterior probabilities and determine which hypothesis is preferred.

(OR)

2. Use Naïve Bayes algorithm to determine whether a "Red Domestic SUV" Car is stolen or not. [6M]

No.	Colour	Type	Origin	Stolen
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Yellow	SUV	Imported	Yes
4	Yellow	SUV	Domestic	No
5	Red	Sports	Imported	Yes

3. Apply the Sequential Covering Algorithm on the dataset below and derive the rule set [6M]

Weather	Wind	Play
Sunny	Weak	Yes
Sunny	Strong	No
Rain	Weak	Yes
Rain	Strong	No
Overcast	Weak	Yes

(OR)

4. A learning algorithm is applied to a Boolean concept learning problem with 4 attributes.

   a) Compute the total number of hypotheses in its finite hypothesis space.

   b) Calculate the maximum possible number of mistakes (mistake bound) before convergence using Conjunction Learning Algorithm